

ERGMs (*ergums*): Statistical Models for Social Networks

- Recent “paradigm shift” in network modeling
 - From “indices” to p -values (let’s get published!)
- Promise to finally get at *the generative processes that govern tie formation in networks*
- From static “**pictures of structure**” to a window on “**generative dynamics**.”
- A way to model how **local structure** gives rise to **global structure**
- Allows us to (finally) see observed relational data as a (stochastic) *realization* of some *underlying data generating process* (DGP)
 - A useful way to look at *all* data not just “network” or “relational” data.

ERGMs (*ergums*): Statistical Models for Social Networks

- The basic insight: breaking free from *The Matrix*.
- Thought you had a “network”?
 - Think again. You actually just have a **binary response variable**.
 - The variable is just presented in a funky format (a $g \times g$ **matrix**). So Just “string the matrix out” into a $[g(g-1)] \times 1$ **vector**.”
 - Does it look like a nail now? Then you know what to do: use the sociologist’s hammer on it.

The Hammer

$$\Pr(\mathbf{Y} = \mathbf{y}) = \left(\frac{1}{\kappa} \right) \exp \left\{ \sum_A \eta_A g_A(\mathbf{y}) \right\}$$

ERGMs (*ergums*): Statistical Models for Social Networks

- Constructing an “ergum”
 - The dependent variable is the absence or presence of a tie.
- Two classes of predictor variables:
 - Dyad independent (or intra-dyadic) processes
 - Triadic balance
 - Dyad dependent (or inter-dyadic) processes
 - Reciprocity, Homophily

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- Models that presume that the network is generated through intra-dyadic processes can be built “the dumb way:”
 - Run a logistic regression on the strung out network with change-score indicators (e.g. “dummy variables”) for the relevant configuration of interest as predictors.
 - The classic Erdos-Renyi “Bernoulli” model is a dyad independent
 - The classic Holland-Leinhardt “p1 model” is a (more interesting) dyad independent model.

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Edge formation probability

$$\text{logit } P(x_{ij} = 1 | G^{-ij}) = \theta$$

Expansiveness

$$\text{logit } P(x_{ij} = 1 | G^{-ij}) = \theta + \rho x_{ji} + \alpha_i + \beta_j .$$

Mutuality

Popularity

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- Models that presume that the network is generated through intra-dyadic and *inter*-dyadic processes have to built the “smart” way.
 - Markov Chain Monte Carlo (MCMC) Estimation.
 - Why the complication?

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- Predictors are direct functions of the response variable (e. g. the existence of a triadic or higher order structure is dependent on the $i \rightarrow j$ link).
- Makes this class of ERGMs an **autoregressive** model.

$$\Pr(\mathbf{Y} = \mathbf{y}) = \left(\frac{1}{K} \right) \exp \left\{ \sum_A \eta_A g_A(\mathbf{y}) \right\}$$

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- These models are somewhat cumbersome to build, because the specification of each “variable” requires some sort of cycling through the network.

$$\text{prob}(\mathbf{X} = \mathbf{x}) = (1/\kappa) \exp \{ \theta L(\mathbf{x}) + \sigma_2 S_2(\mathbf{x}) + \sigma_3 S_3(\mathbf{x}) + \tau T(\mathbf{x}) \}, \quad (3)$$

